BEYOND TECHNOLOGY

Why Prosperity Is Built, Not Downloaded

What Two Centuries of Data Reveal About Africa's Economic Future

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Executive Summary

Countries become prosperous through industrialization and structural transformation: the sustained shift of labor and capital from low-productivity activity into higher-productivity manufacturing, agro-processing, and tradable services. Successful transformation depends on the accumulation of multiple complementary capabilities, including education, health, agricultural productivity, energy, infrastructure, capital formation, development finance, and institutions. Technology contributes to this process when it strengthens these capabilities. This report's central finding, drawn from two centuries of comparative data, is that technology has not historically been sufficient to produce broad-based prosperity on its own, in any of the cases examined.

Prosperity, as this report uses the term, is broader than startup funding, mobile-phone adoption, GDP growth, or digital connectivity. It is the sustained expansion of productive capabilities across an entire economy: rising incomes, productive employment, industrial capability, food security, reliable infrastructure, technological advancement, and improving living standards across broad sections of society, not in a single sector or city.

This is a comparative and historical report, not a causal-identification study. Where the text states that one factor caused another, it does so only where the cited literature supports a causal claim; elsewhere, it describes association, sequence, and plausible mechanism, and says so explicitly. A dedicated note on method describes this distinction in full.

Key findings

Africa's starting deficits predate the smartphone by a century. Adult literacy in Sub-Saharan Africa (~68%) remains below Western Europe's pre-1950 level; life expectancy (62.7 years) trails the global average by over a decade; electricity access (53%) trails the 92% global average; and per-capita electricity consumption outside South Africa is a small fraction of the global norm.

Large income gaps have closed within one to two generations, but not through technology alone. Historical catch-up cases (principally South Korea and China, and a wider panel including Vietnam, Poland, Malaysia, and Costa Rica) closed large income gaps through the sustained, simultaneous accumulation of literacy, health, agricultural productivity, capital, and tradable production capacity. In every large economy, that capacity was built on manufacturing; services led only in small economies, and even there on an industrial base. No case in this report achieved sustained catch-up primarily through technology leapfrogging.

Manufacturing remains the only path that has worked at scale. Manufacturing has historically been the sector most able to absorb low-skill labor at rising productivity, and Sub-Saharan Africa's manufacturing value added (~10% of GDP) is roughly two-fifths of China's and South Korea's (~25% each). This report also engages directly with the strongest challenges to a manufacturing-first strategy: automation is compressing the labor-intensive manufacturing window East Asia exploited; India has pursued productivity growth through knowledge-intensive services instead, though its income per person remains less than half of China's; and global trade is fragmenting in ways that may

narrow, though not eliminate, the export-led manufacturing path. The report's working principle is that productive complexity matters more than sector labels: modern industrialization can include manufacturing, agro-processing, digital and business services, and logistics, provided each raises productivity and absorbs labor at scale.

Africa is not a single economy. Fragile low-income states, diversified middle-income economies, resource exporters, emerging manufacturing hubs, and governance reform leaders face materially different constraints. The continental averages used throughout this report should be read as a starting frame, not a substitute for country-level analysis.

Institutions matter, but causation runs both ways. Institutions and state capacity correlate strongly with prosperity, but the direction of causation is genuinely contested in the literature: income and institutional quality are believed to reinforce each other over time (Besley and Persson, 2011), and industrial policy has failed at least as often as it has succeeded, including in Africa's own 1960s–1980s import-substitution experience. Where South Korea and China succeeded, capable, continuity-oriented states appear to have been a necessary, not sufficient, condition.

Continuity must be legislated where it cannot be assumed. South Korea and China took long policy horizons for granted under directive states. In African democracies, where a change of government often reverses development plans (Nigeria counted 11,886 abandoned public projects in 2011), multi-year development budgets on the model of the EU's seven-year financial framework offer a democratic route to the same continuity.

Integrity is a development capability. Long-horizon budgets are also larger targets for theft. Ring-fenced, results-based, independently monitored disbursement, as used for the Abacha funds Switzerland returned to Nigeria in 2017, is a condition of development spending, not an afterthought.

Resource dependence is not destiny. Whether natural-resource wealth helps or harms an economy is conditional on institutional quality, as Botswana's diamond-financed growth illustrates against Nigeria's and Angola's oil-financed stagnation (Mehlum, Moene, and Torvik, 2006).

Policy implications. Governments face a narrower and more contested version of the sequencing this report describes than South Korea and China faced. Literacy, health, and agricultural productivity remain close to unconditional priorities across the literature; the manufacturing-versus-services and state-led-versus-market-led questions are genuinely open and country-specific. Each recommendation in the policy blueprint is stated with the strength of evidence behind it, not as a uniform prescription.

Conclusion. No country in this report became prosperous through technology alone, education alone, infrastructure alone, or manufacturing alone. Prosperity emerged, in every case, once a sufficient threshold of complementary capabilities had accumulated and begun reinforcing one another. Good governance (capable public administration, credible property rights, and policy continuity) is the capability most consistently associated with a country's ability to build and sustain the rest. Technology strengthens that process. The evidence does not support the claim that it can replace it: software does not build a productive base, and in every case examined here, that base came before

technology could multiply it.

Introduction

Nigeria is this report's motivating puzzle. The country has one of Africa's most vibrant technology ecosystems: globally recognized fintech firms, mobile-phone penetration near the continental norm, and substantial entrepreneurial energy. Yet manufacturing remains limited, electricity is unreliable enough that firms routinely self-generate power at a cost premium, logistics costs are high, agricultural productivity is low even by regional standards, learning outcomes are weak, and productivity growth has been uneven across the economy. Nigeria's rule-of-law and governance indicators sit below those of several lower-income regional peers (Section 9), and its oil-dependent growth has not converted resource revenue into broad-based prosperity as consistently as Botswana's diamond-financed growth has (Section 12).

The puzzle is not that Nigeria's technology sector has failed. It is that a genuinely competitive technology sector has not, on its own, been sufficient to produce the broad-based prosperity the rest of the economy would need to converge with global incomes. This report argues that Nigeria's experience is not an exception to explain away but a clear illustration of its central proposition: prosperity emerges when multiple complementary capabilities develop together, not when any single capability, however impressive, develops in isolation.

That proposition, generalized, is this report's organizing principle: countries achieve prosperity through industrialization and structural transformation, and successful transformation depends on the accumulation of complementary capabilities, including education, health, agriculture, infrastructure, energy, capital formation, institutions, and technology. Technology contributes to prosperity when it strengthens these foundations. It has not, in any case examined here, been sufficient to create broad-based prosperity on its own.

Defining prosperity. This report defines prosperity as a condition characterized by rising incomes, productive employment, industrial capability, food security, reliable infrastructure, technological advancement, and improving living standards across broad sections of society. That is a deliberately broader definition than startup funding, mobile-phone adoption, GDP growth alone, or digital connectivity, each of which can rise in a capital city or a single sector while the underlying economy remains largely unchanged. Prosperity, in the sense this report uses the term, is the sustained expansion of productive capabilities across an entire economy.

The FutureLabs Prosperity Framework. This report introduces the FutureLabs Prosperity Framework, an organizing structure rather than a new empirical claim. It proposes that countries become prosperous not by maximizing a single variable but by building a broad base of complementary capabilities that support industrialization, productivity growth, and structural transformation. The framework comprises education and skills, health and nutrition, agricultural productivity, reliable energy, infrastructure and logistics, capital formation, development finance, manufacturing and tradable services, and institutions and state capability. Technology and

innovation sit across all of these as a cross-cutting capability that strengthens each of the others, rather than as a separate, additive layer. Mobile phones do not replace electricity; fintech does not replace industry; artificial intelligence does not replace education; startups do not replace infrastructure. Digital adoption does not replace productive transformation. Section 13 presents the framework in full, including the evidence basis for each capability's inclusion.

The threshold concept. A recurring pattern across the historical cases in this report is that countries become prosperous only after accumulating a sufficient threshold of complementary capabilities, not after maximizing any one of them. No country examined here became prosperous through technology alone, education alone, infrastructure alone, or manufacturing alone. In each successful case, prosperity emerged once enough capabilities had developed far enough, together, to reinforce one another: literate and healthy workers made capital investment productive, reliable power made manufacturing viable, and capable institutions made both durable. This report treats that threshold, rather than any single input, as the more useful unit of policy analysis, and returns to it directly in the policy blueprint.

This report is organized in thirteen sections. It begins by situating Africa's considerable internal diversity and introducing the wider set of comparator economies used throughout. It then examines, in turn, human capital, agriculture, energy, capital formation and its financing, manufacturing and its principal alternatives and constraints, institutions, state capability, budget continuity and integrity, demographics, and technology as a cross-cutting capability, before closing with a wider comparative panel and the full Prosperity Framework and policy blueprint. A note on method below, and a closing section on methodology and limitations, state explicitly what kind of evidence supports each claim and where that evidence is contested.

A Note on Method and Evidence

This report draws on four distinct types of evidence, and the strength of each claim should be read accordingly.

Descriptive evidence (the majority of the report's charts and cross-country comparisons) establishes what the current gaps are: how Sub-Saharan Africa's literacy, electrification, or manufacturing share compares with other regions today. These comparisons are reliable as description but do not, by themselves, establish why the gaps exist.

Historical comparison (the South Korea, China, and wider-panel case material, including Nigeria's own experience) establishes what sequence of investments preceded income convergence in specific countries. This evidence is suggestive of mechanism but is drawn from a small number of cases selected partly because they succeeded, a limitation discussed further below.

Causal inference, in the stricter econometric sense, is used sparingly in this report and is flagged explicitly where it appears: for instance, the quasi-experimental literature on rural electrification's effect on employment, or Mehlum, Moene, and Torvik's (2006) cross-country evidence that the resource curse is conditional on institutional type. Most of the relationships described in this report,

including the correlation between rule of law and income and between tax capacity and public investment, have not been causally identified in the underlying literature and are presented as robust associations with plausible, contested mechanisms.

Policy interpretation is this report's own synthesis, presented in the concluding blueprint, and is explicitly normative: it reflects a judgment about which investments the descriptive and historical evidence most consistently supports, not a claim that the evidence uniquely determines this policy path.

1. Africa in Comparative Context: Heterogeneity and the Comparator Set

Continental and Sub-Saharan averages, used throughout this report for tractability, obscure differences among African economies at least as large as the differences between Africa and the rest of the world. Five broad groups are useful for interpreting the country-level exceptions that recur throughout this report:

- **Low-income fragile states** (South Sudan, Somalia, the Central African Republic, Chad, Sahelian states affected by conflict) face binding constraints on nearly every dimension this report examines simultaneously, and are unlikely to follow any single-sector sequencing recommendation without a prior resolution of security and basic state functioning.
- **Diversified middle-income economies** (South Africa, Kenya, Egypt, Morocco) already have partial manufacturing and services bases and higher literacy and life expectancy than the continental average, and are the most plausible near-term candidates for the manufacturing- or services-upgrading strategies discussed later in this report.
- **Resource exporters** (Nigeria, Angola, Zambia, the Democratic Republic of Congo) generate substantial export revenue from oil, gas, or minerals but have historically converted that revenue into broad-based prosperity inconsistently; Section 12 examines why, using Botswana as a counter-example within the same resource-exporter category.
- **Emerging manufacturing hubs** (Ethiopia, Rwanda, Morocco, increasingly Egypt) have pursued export-processing-zone and manufacturing-FDI strategies with mixed but generally more encouraging results than the continental average, discussed in Section 8.
- **Governance reform leaders** (Rwanda, Mauritius, Botswana, increasingly Ghana) score well above the continental average on institutional and rule-of-law measures relative to income, and are the closest available evidence, within Africa, for what the institutional argument in Section 9 predicts.

A wider comparator set. South Korea and China, this report's principal historical cases, are extreme rather than typical examples of catch-up growth, and the report draws throughout on a wider panel to test how far their lessons generalize. The panel includes Vietnam, Bangladesh, Poland, and Romania, along with six further cases used at relevant points in the sections that follow: Malaysia and

Indonesia, resource-rich Southeast Asian economies that industrialized without abandoning commodity exports; Costa Rica, a small economy that reached upper-middle income through knowledge-intensive services built on an electronics-manufacturing anchor rather than through mass manufacturing; Chile, a resource exporter whose institutional quality is the leading explanation for its divergence from other mineral-dependent Latin American economies; Turkey, a manufacturing-led upper-middle-income economy whose recent institutional backsliding complicates any strict claim that institutional quality and growth move together; and Singapore, a city-state whose extreme technocratic state capacity produced convergence within a single generation but whose small scale and unique geography limit how much of its experience transfers to a continent of 1.5 billion people across 54 states. Table 1 summarizes where each comparator recurs.

Comparator	Primary relevance	Section
South Korea, China	Core sequencing cases	Throughout
Vietnam, Bangladesh, Poland, Romania	Wider catch-up panel; manufacturing diversification vs. single-sector risk	12
Malaysia, Indonesia	Manufacturing alongside continued resource exports	12
Costa Rica	Small-economy services path on an electronics-manufacturing base; tests how far it scales	8, 12
Chile, Botswana	Resource dependence conditional on institutional quality	12
Turkey	Institutional backsliding alongside continued manufacturing growth	9
Singapore	Extreme state capacity; limited scale transferability	9

Table 1. Comparator economies used in this report and their primary analytical role.

No claim in this report treats any single comparator as a template. Each is used to test whether a specific mechanism (manufacturing-led absorption of labor, institutional causation, or resource-curse conditionality) generalizes beyond the South Korea and China cases, and each section notes where the comparator complicates rather than confirms the report's central argument.

2. Human Capital: Education

Every catch-up case in this report began with a sustained, government-led literacy campaign launched decades before its income effects became visible. By 1900, Western Europe and North America had already passed 90% adult literacy; Sub-Saharan Africa entered the twentieth century near 8% and reached roughly 68% by 2023, a level China reached in the early 1980s.

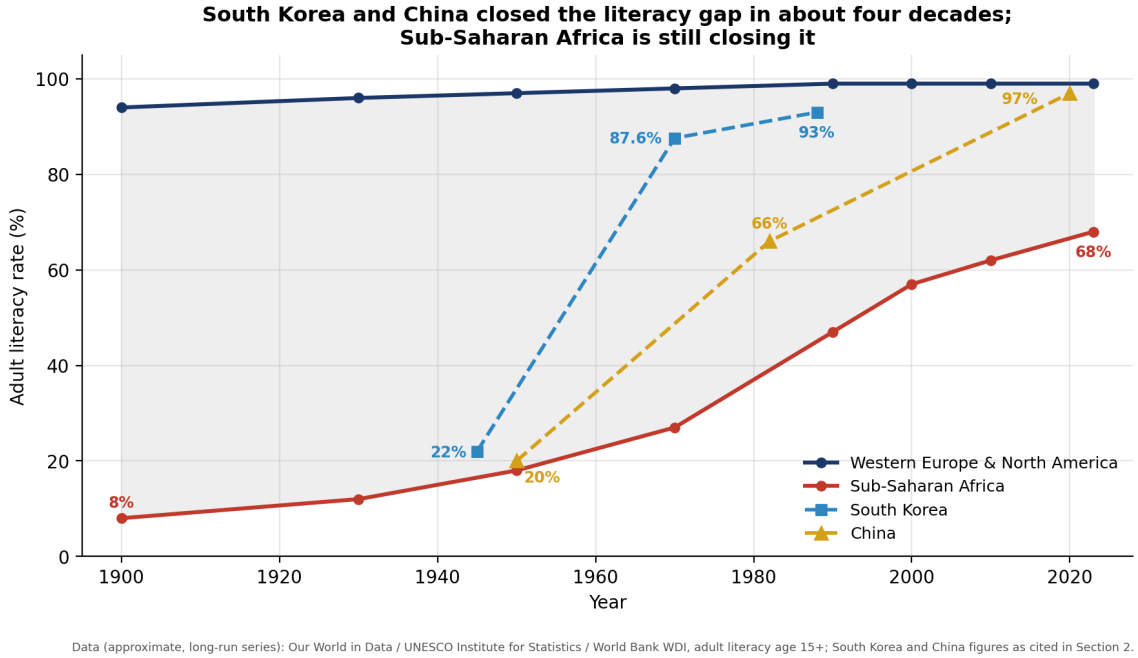


Figure 1. Adult literacy rate (%), Western Europe & North America, Sub-Saharan Africa, South Korea, and China, 1900–2023.

Source: Our World in Data / UNESCO Institute for Statistics / World Bank World Development Indicators, adult literacy age 15+; South Korea and China figures as cited in the text.

South Korea's literacy rose from about 22% in 1945 to 87.6% by 1970 and above 93% by the late 1980s. China's rose from about 20% in 1950 to 66% in 1982, the year nine-year compulsory education was constitutionally entrenched, and to 97% by 2020. Both trajectories illustrate that literacy compounds across generations through household effects on subsequent schooling demand, a mechanism with reasonably strong support in the education-economics literature, though the magnitude of the intergenerational effect varies by setting. Africa's current trajectory resembles these historical cases in shape; what most African states have not yet demonstrated is the four-to-seven-decade policy continuity that let South Korea's and China's campaigns run to completion, a theme this report returns to directly in Section 9.

3. Health: The Second Pillar of Human Capital

Sub-Saharan Africa's life expectancy (62.7 years) trails the global average (73–74 years) by more than a decade; South Korea's (83.4 years) illustrates the distance a health system can travel, having itself been in the low 50s as recently as the early 1960s.

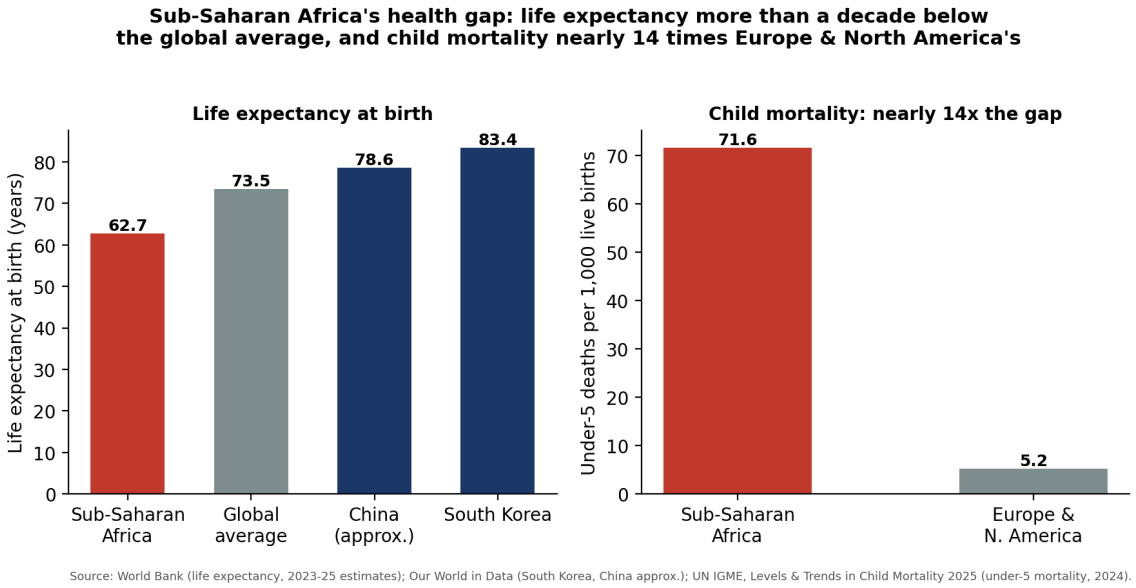


Figure 2. Life expectancy at birth and under-5 mortality, Sub-Saharan Africa vs. global average vs. China vs. South Korea, most recent available data.

Source: World Bank (life expectancy, 2023–25 estimates); Our World in Data (South Korea, China approximate figures); WHO/UNICEF/UN Inter-agency Group for Child Mortality Estimation, Levels & Trends in Child Mortality 2025 (under-5 mortality, 2024).

Under-5 mortality (71.6 per 1,000 live births) is roughly fourteen times the Europe and North America rate, driven substantially by preventable or treatable causes. The financing gap behind this is stark: Sub-Saharan Africa spends roughly $92 per person on health against $672 in China and a $1,325 global average.

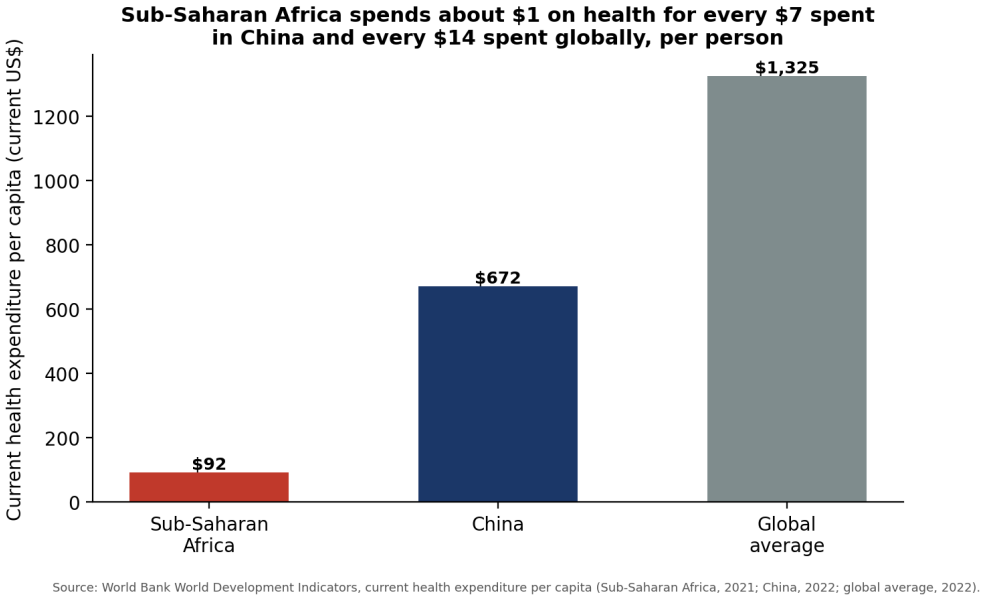


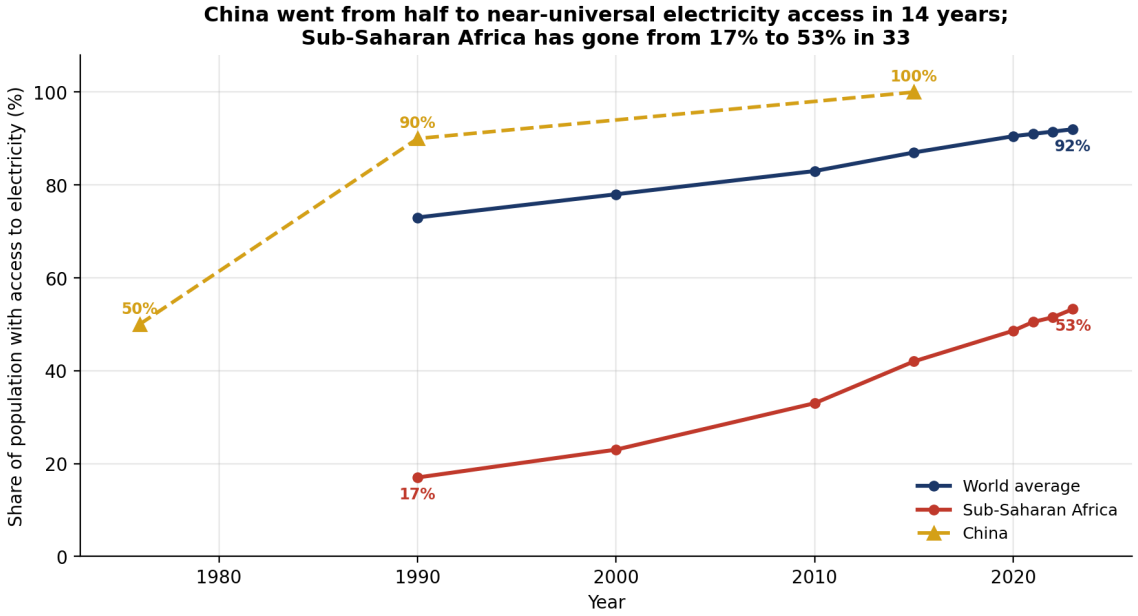
Figure 3. Current health expenditure per capita (current US$), Sub-Saharan Africa vs. China vs. global average, most recent available year.

Source: World Bank World Development Indicators, current health expenditure per capita (Sub-Saharan Africa, 2021; China, 2022; global average, 2022).

The relationship between health and income growth is plausible and widely assumed, but it is worth being precise about what the literature actually shows. Micro-level evidence that healthier individuals earn more and that childhood health affects adult productivity is reasonably robust. The macro-level claim, that raising a country's life expectancy mechanically raises its GDP growth, is more contested: Acemoglu and Johnson's influential 2007 study of the international epidemiological transition found that large, plausibly exogenous gains in life expectancy after 1940 raised population and labor supply substantially but had little discernible effect on GDP per capita over the following decades, because a larger population diluted the gain across more people. This report treats health investment as a component of human capital and an end in itself on welfare grounds, and as plausibly complementary to the other investments it examines, without claiming the macro growth effect of health improvements alone is as well established as its effect on individual welfare and productivity.

4. Energy

Sub-Saharan Africa's electricity access (53%) trails the global average (92%); roughly 565 million people in the region lacked electricity in 2023. China, by contrast, moved from about half to 90% access between 1976 and 1990 and reached universal access in 2015.

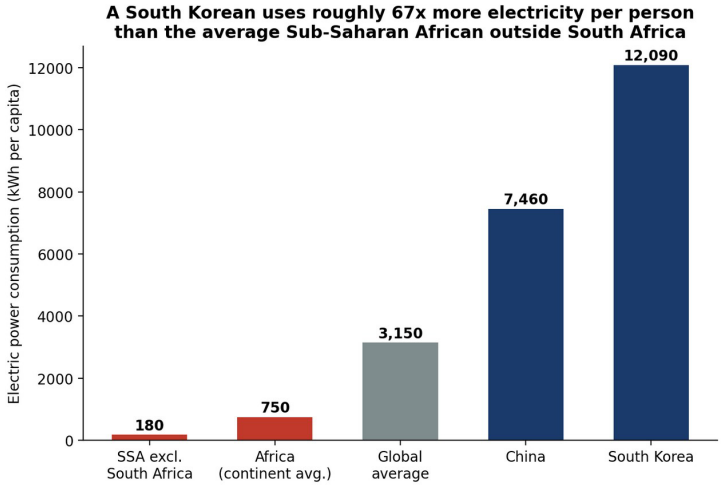


Source: World Bank / ESMAP, Tracking SDG7 (2024); World Bank WDI. China 1976-1990 rural electrification as cited in Section 4; universal access reached in 2015.

Figure 4. Share of population with access to electricity (%), World average, Sub-Saharan Africa, and China, 1976–2023.

Source: World Bank / ESMAP, Tracking SDG7: The Energy Progress Report (2024); World Bank World Development Indicators; China's 1976–1990 rural electrification as cited in this section.

Access understates the gap because it does not capture how much power a connection can reliably deliver. Measured by consumption, Africa's continental average is roughly 750 kWh per person, and roughly 180 kWh outside South Africa, against 7,460 kWh in China and 12,090 kWh in South Korea.



Source: BloombergNEF/World Bank, Africa Power Transition Factbook 2024 (Africa, SSA excl. South Africa, global, 2023); Ember/Our World in Data, via Wikipedia (China, South Korea, 2025).

Figure 5. Electric power consumption per capita (kWh), selected regions and countries, most recent available year.

Source: African Development Bank; BloombergNEF/World Bank, Africa Power Transition Factbook 2024 (Africa, SSA excl. South Africa, global figures, 2023); Ember/Our World in Data compilation (China, South Korea, 2025).

A broader cross-country sample confirms the pattern holds beyond the handful of economies above: electricity consumption per person and GDP per capita rise together across a wide range of income levels, and no country in the sample has reached high income while remaining a low-electricity economy.

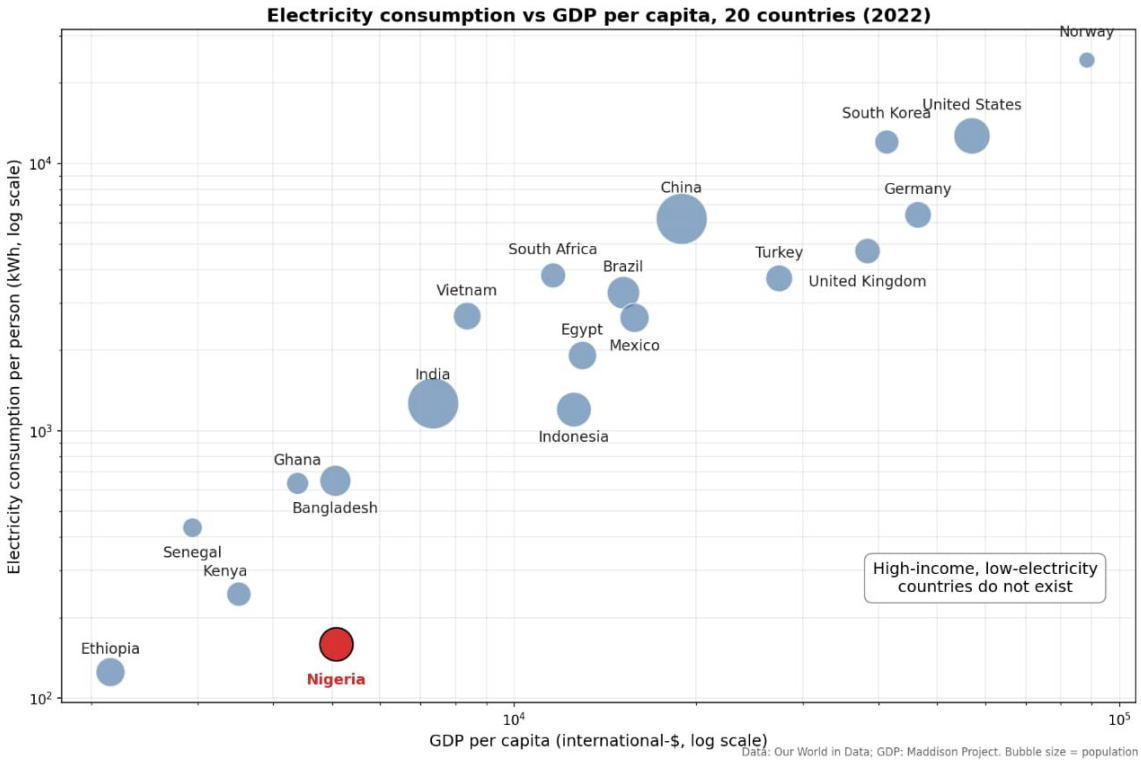


Figure 6. Electricity consumption vs. GDP per capita, 20 countries (2022), bubble size = population.
Source: Our World in Data (electricity consumption); Maddison Project (GDP per capita). Bubble size = population.

The direction of causation between electrification and income cannot be established from cross-sectional data of this kind alone; wealthier countries can afford more power generation just as plausibly as more power generation enables wealth, and the two almost certainly reinforce each other over time. The stronger causal evidence comes from a different literature: randomized and quasi-experimental studies of specific grid-extension programs (for example, rural electrification in India, South Africa, and Brazil) generally find modest but real effects on local employment and firm formation, concentrated in areas with pre-existing commercial activity, and smaller effects on household income alone. That literature supports a narrower claim than the cross-country scatter implies: electrification appears to enable productive activity that is already latent, rather than mechanically generating growth wherever it arrives. This is consistent with this report's broader argument that infrastructure functions as a complement to other inputs rather than a sufficient condition on its own. China's rural electrification (roughly 50% to 90% access, 1976–1990) was explicitly paired with rural industrial policy rather than deployed as a standalone program, consistent with this complementarity.

5. Agriculture: Productivity Before Transformation

Labor released into industry or services has, in every historical case examined here, come predominantly from agriculture, which makes agricultural productivity, not agricultural output, a precondition for the sectoral transitions discussed later in this report.

Sub-Saharan Africa's cereal yield (~1,650 kg/hectare) is roughly a quarter of China's (6,380 kg/hectare) or South Korea's (~6,300–6,800 kg/hectare), and below the South Asian average (~3,705 kg/hectare).

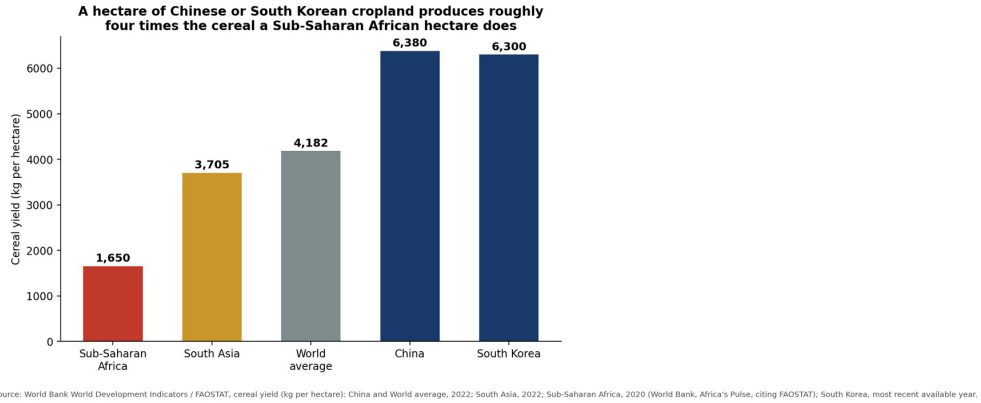


Figure 7. Cereal yield (kg per hectare), Sub-Saharan Africa vs. South Asia vs. World average vs. China vs. South Korea, most recent available year.

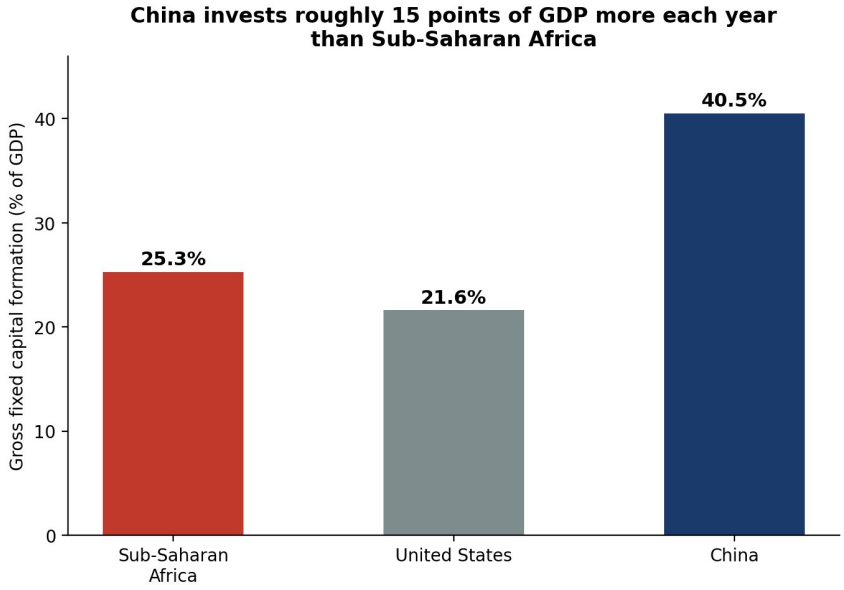
Source: World Bank World Development Indicators / FAOSTAT, cereal yield (kg per hectare): China and World average, 2022; South Asia, 2022; Sub-Saharan Africa, 2020 (World Bank, Africa's Pulse, citing FAOSTAT); South Korea, most recent available year.

World Bank research attributes most of this gap to extensive rather than intensive growth: African agricultural output has expanded primarily through cultivating more land rather than through the fertilizer use, irrigation, and improved seed varieties that raised yields elsewhere. Total factor productivity in the region's agriculture grew at 0.2% annually between 2000 and 2019, against 1.8–2.5% in Latin America and Asia, and fertilizer application remains well below the 50 kg/hectare benchmark commonly used as a reasonable minimum.

South Korea's and China's own transitions both began with a state-directed push to raise farm yields (land reform and input-intensive cultivation in Korea's case, the Household Responsibility System from 1978 in China's) specifically to secure urban food supply and free labor before or alongside the manufacturing buildout described in Section 8. Raising Sub-Saharan African yields toward the South Asian benchmark would, on this evidence, free a meaningful share of the region's rural workforce for the non-farm employment this report's later sections identify as the central labor-market challenge of the coming decades.

6. Investment Builds Nations

Gross fixed capital formation, roughly 25.3% of GDP in Sub-Saharan Africa and comparable to the United States (21.6%), sits well below China's sustained 40.5%.



Source: World Bank World Development Indicators, Gross fixed capital formation (% of GDP), 2023 (China, Sub-Saharan Africa) / 2024 (United States).

Figure 8. Gross fixed capital formation (% of GDP), Sub-Saharan Africa vs. United States vs. China, most recent available year.

Source: World Bank World Development Indicators, Gross fixed capital formation (% of GDP): China and Sub-Saharan Africa, 2023; United States, 2024.

Because capital stock compounds, a sustained fifteen-point gap in annual investment widens over time rather than closing on its own. This is also a structural limit on leapfrogging: a port still has to be dredged, a factory built and equipped, a road paved. Digital tools can lower the transaction cost of financing investment (mobile-money savings mobilization, more transparent procurement) but do not substitute for the underlying capital expenditure.

The more consequential question, addressed next, is not simply how much capital formation occurs but how it is financed, since the composition of financing determines whether a given investment rate is sustainable.

7. Financing Development

Sub-Saharan Africa's public debt rose from roughly 30% of GDP (2013) to almost 60% (2024). External debt service consumed 18.7% of government revenue in 2024, roughly three times the 2014 share, and total external debt surpassed $1 trillion, more than double its 2020 level.

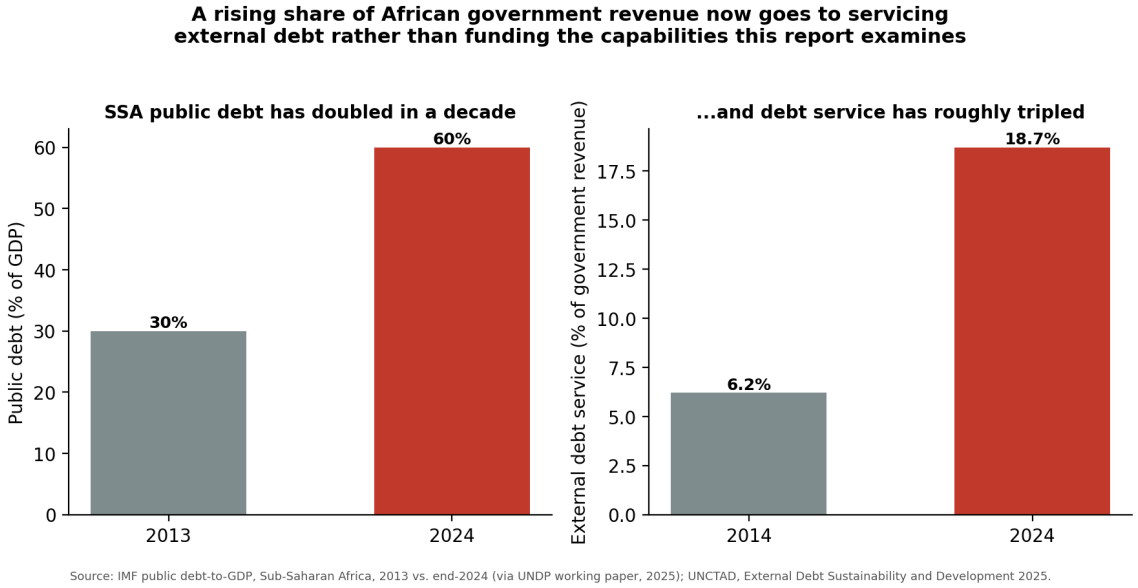


Figure 9. Sub-Saharan Africa's public debt (% of GDP) and external debt service (% of government revenue), 2013/2014 vs. 2024.

Source: IMF public debt-to-GDP data for Sub-Saharan Africa, 2013 vs. end-2024, via UNDP working paper (2025); UNCTAD, External Debt Sustainability and Development 2025.

Every dollar spent on debt service is unavailable for the investments discussed elsewhere in this report. The World Bank estimates roughly $20 billion in external public and publicly guaranteed interest payments across the region in 2025, close to three-quarters owed to private creditors and Chinese official and private lenders. Concessional aid is a shrinking offset: the ISS African Futures programme projects official development assistance falling from 2.4% of continental GDP (2023) to roughly 1.6% by 2043, while OECD Development Assistance Committee disbursements fell 7.3% in 2024 alone.

Foreign direct investment, equity-like capital with no fixed repayment schedule, remains thin: Africa's FDI inflows averaged roughly 2% of GDP (1990–2022) and are concentrated in five countries. China's FDI inflows reached nearly 5% of GDP in the late 1990s, roughly two-thirds directed to manufacturing. South Korea's foreign capital inflows in its comparable period, close to 10% of GDP in the early 1970s, came overwhelmingly through long-term loans and trade credit under capital controls rather than equity, but were directed deliberately at export-manufacturing capacity rather than general budget financing.

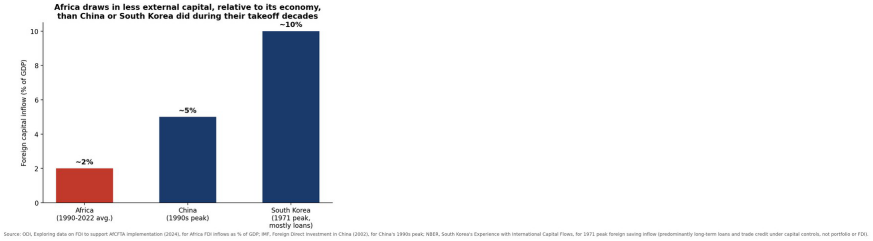


Figure 10. Foreign capital inflows (% of GDP), Africa (1990–2022 average) vs. China (1990s peak) vs. South Korea (1971 peak, predominantly loans), for comparison.

Source: ODI, Exploring data on FDI to support AfCFTA implementation (2024); IMF, Foreign Direct Investment in China (2002); NBER, South Korea's Experience with International Capital Flows.

The comparison suggests, though it does not conclusively prove given the differing eras and data conventions involved, a financing-composition principle distinct from the volume principle above: capital directed toward productive, tradable-sector investment tends to generate the export earnings needed to service itself, while debt financing general consumption raises short-run capital-formation figures without building that repayment capacity. A given capital-formation rate financed by productive investment is not the same achievement, durability considered, as the same rate financed by short-tenor external debt.

8. Manufacturing, Its Alternatives, and Its Limits

Manufacturing value added is roughly 10% of GDP in Sub-Saharan Africa against roughly 25% in China and South Korea.

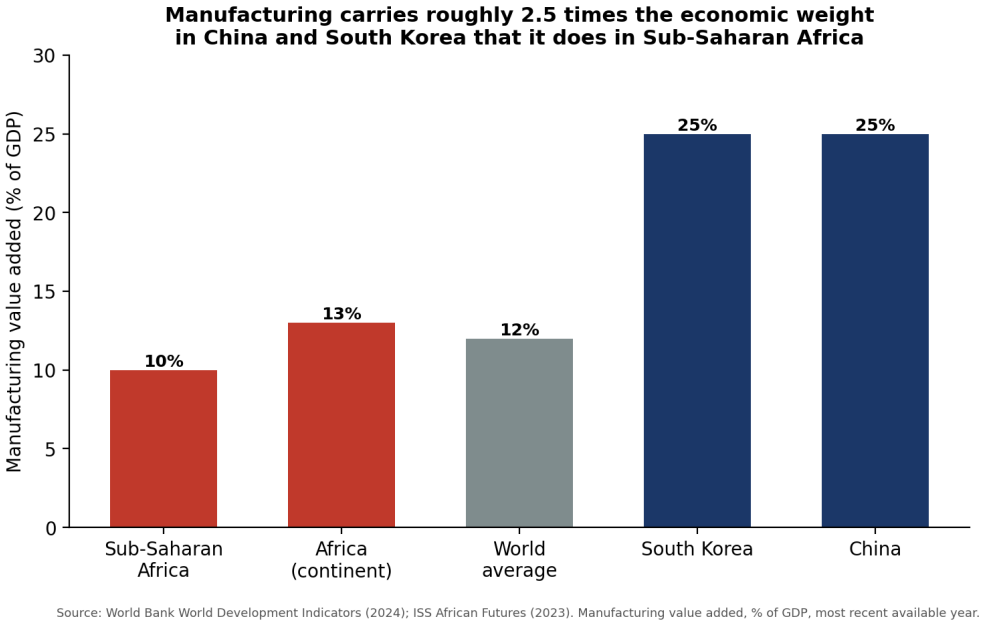


Figure 11. Manufacturing value added (% of GDP), Sub-Saharan Africa vs. Africa (continent) vs. World average vs. South Korea vs. China, most recent available year.

Source: World Bank World Development Indicators (2024); ISS African Futures, Africa Manufacturing Forecast (2023 continental figure); World Bank data (China).

The traditional case for manufacturing's centrality, associated with Arthur Lewis and, in its modern form, with Dani Rodrik, rests on three linked claims: it absorbs low-skill labor directly out of agriculture; the productivity gains from that absorption compound over time as firms move up the value chain; and wages track productivity closely enough that this compounding shows up as sustained real-wage growth. Complementary indicators, such as low manufactured-export shares and Economic Complexity Index rankings for most of Sub-Saharan Africa well below China's and dramatically below South Korea's, are consistent with this pathway remaining largely unrealized on the continent.

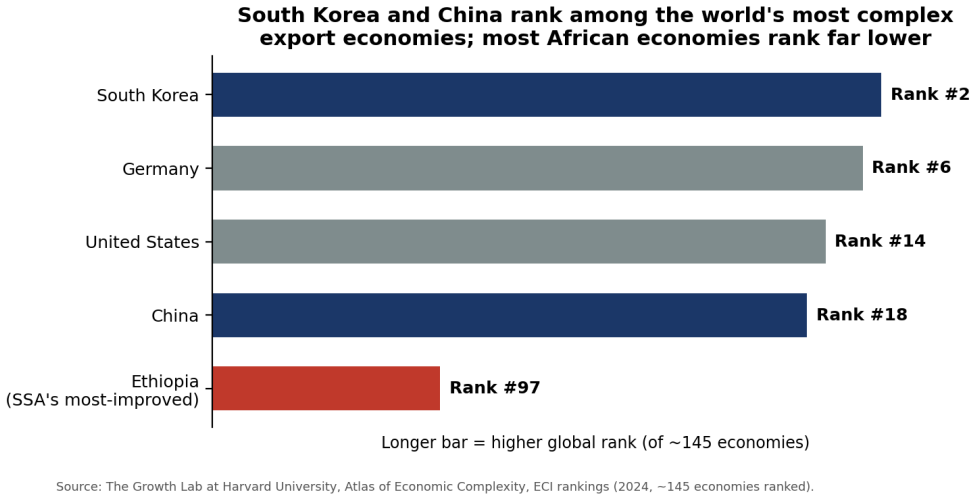


Figure 12. Economic Complexity Index global ranking, selected countries, 2024.

Source: The Growth Lab at Harvard University, Atlas of Economic Complexity, ECI rankings (2024, roughly 145 countries ranked).

The strongest counterargument: automation and premature deindustrialization. Rodrik's own subsequent work is also the strongest challenge to a manufacturing-first strategy. Western European countries reached peak industrialization at roughly $14,000 per capita (1990 dollars); India and most of Sub-Saharan Africa appear to be reaching peak manufacturing employment share at incomes closer to $700, a pattern Rodrik terms premature deindustrialization. Recent research (Kunst, cited in Center for Global Development analysis) attributes a substantial share of this pattern specifically to automation displacing the unskilled manufacturing jobs developing countries would previously have absorbed, rather than to trade competition alone, though the relative weight of the two explanations remains actively debated. The evidence is not one-directional: China's own experience shows manufacturing employment holding roughly steady through the 2010s and 2020s even as its industrial robot stock grew tenfold (2014–2024), suggesting large-scale labor absorption can coexist with rapid automation, at least at China's scale and stage of development. The balance of evidence supports a narrower conclusion than either the optimistic or pessimistic extremes: the window for labor-intensive manufacturing-led catch-up has narrowed relative to South Korea's and China's era, without having closed, and countries entering it later face a harder version of the same strategy rather than a foreclosed one.

A genuine alternative: services-led development. A substantial and growing literature, including recent World Bank research (Hallward-Driemeier and colleagues) and Rogerson, Herrendorf, and Valentinyi's structural-transformation work, argues that knowledge-intensive, globally tradable services (IT, business-process outsourcing, professional and technical services) can play a role comparable to manufacturing's historical one. India's IT-led growth is the leading example, and in Costa Rica, Ghana, India, Pakistan, and the Philippines, IT and professional services already account for more than half of services exports. The skeptical case, associated with Rodrik, is that these tradable services remain skill-intensive and absorb only a small share of a typically low-skilled labor force; most services growth in developing economies still occurs in low-productivity, non-tradable

activity (retail, informal transport) that does not replicate manufacturing's historical role at scale. Recent Africa-specific firm-level research (Diao, Ellis, McMillan, and Rodrik, 2025, examining Tanzanian and Ethiopian manufacturers) finds that African manufacturing firms which do achieve productivity gains often fail to scale employment alongside them. That pattern complicates the simple manufacturing-absorbs-labor mechanism this report's earlier sections describe, and suggests the labor-absorption channel may require more deliberate firm-level policy support than historical East Asian cases needed. Set against China, however, India's services-led path looks less like an alternative than a slower route. In 1990, India's income per person was slightly higher than China's (about $368 against $318 in current US dollars, World Bank); by 2024, India's GDP per capita was about $11,000 (PPP) against roughly $24,000 for China, which industrialized first. Costa Rica, now richer per person than China at about $27,000 (PPP) in 2024, is the case that tests the rule rather than breaking it. It is a country of about five million people that built literacy, health, and stable institutions early, and its services boom followed an electronics-manufacturing anchor, Intel's chip-assembly plant, opened in the late 1990s. Costa Rica shows that a small economy can climb through high-value niches on a productive base; it does not show that a large, young population can be absorbed without industry.

Premature deindustrialization within Africa. Zambia, Nigeria, and Ghana saw manufacturing's GDP share rise through the 1990s–2000s before plateauing, often alongside commodity booms that made resource extraction more immediately profitable, consistent with the resource-curse dynamics examined in Section 12. Ethiopia and Rwanda have pursued more deliberate export-zone strategies since roughly 2010 with mixed but generally more encouraging results.

Taken together, the evidence supports treating manufacturing as the only pathway that has delivered labor-absorbing productivity growth at the scale of a large population, with tradable services as a complement rather than a substitute. The working principle this report draws from the debate above is that productive complexity matters more than sector labels: modern industrialization can include manufacturing, agro-processing, digital and business services, and logistics, provided the activity raises productivity and absorbs labor at scale, rather than remaining confined to a narrow, high-skill segment of the economy. The policy blueprint in Section 13 reflects this by recommending that manufacturing strategy be paired with, not treated as a substitute for, deliberate investment in tradable services capacity where a country's skill base and comparative advantage make that pairing viable.

9. Institutions, State Capacity, Continuity, and Integrity

Countries with stronger rule of law cluster at higher income; Nigeria scores low relative to regional peers, a pattern the governance literature links to weaker private investment.

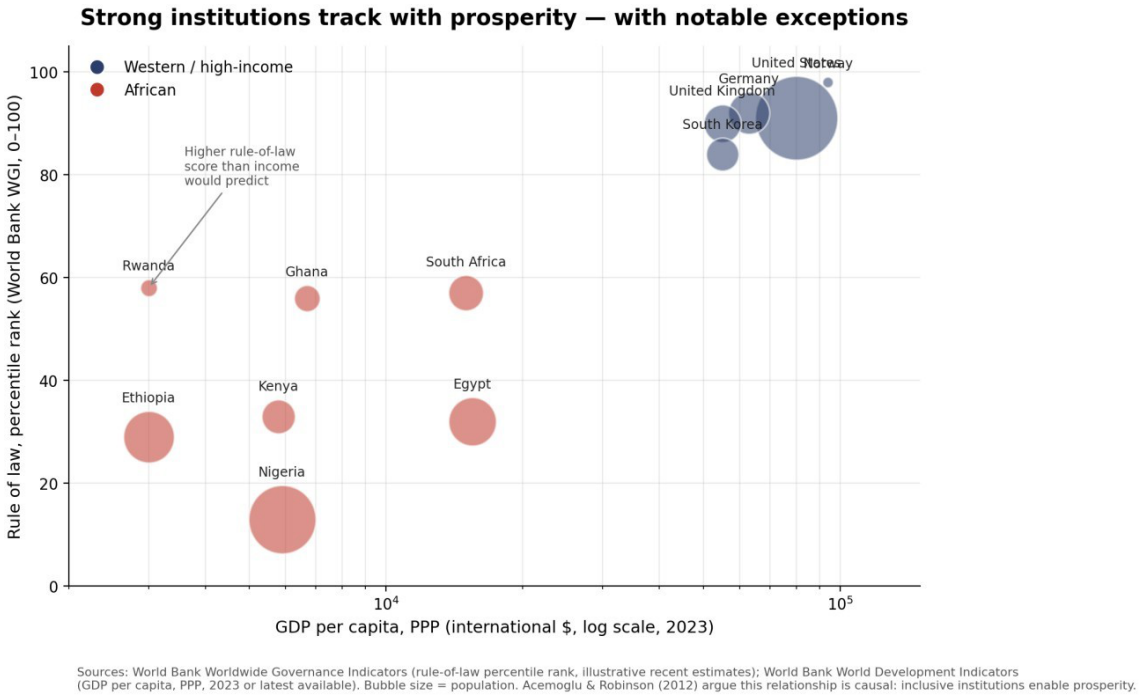


Figure 13. Rule of law (World Bank Worldwide Governance Indicators percentile rank) vs. GDP per capita (PPP, log scale), selected Western and African countries, bubble size = population.

Source: World Bank Worldwide Governance Indicators; World Bank World Development Indicators (2023 or latest available).

Acemoglu and Robinson (2012) argue this relationship is causal: inclusive institutions enable prosperity by allowing individuals to keep the returns on their effort. This is an influential but not a settled claim. Besley and Persson's (2011) formal treatment of state capacity finds the relationship runs in both directions: better legal and fiscal institutions raise growth, but higher income also raises a state's incentive and ability to invest in institutional capacity, meaning cross-sectional correlations of the kind in Figure 13 cannot, on their own, distinguish institutions causing growth from growth financing institutions. Rwanda, whose rule-of-law score exceeds what its income alone would predict, is consistent with (though does not prove) the causal direction Acemoglu and Robinson propose. Turkey, a manufacturing-led upper-middle-income economy whose institutional indicators have weakened over the past decade without a corresponding reversal in per-capita income, is a caution against assuming the relationship is tight or immediate in either direction.

Tax capacity and the sequencing of trust. Africa's tax revenue (16.1% of GDP) trails the OECD (33.9%) substantially.

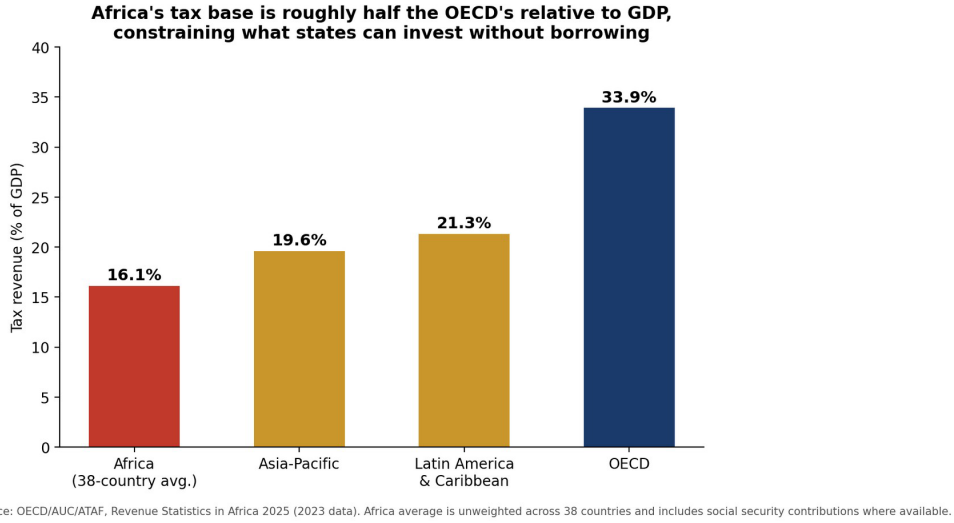


Figure 14. Tax revenue (% of GDP), Africa vs. Asia-Pacific vs. Latin America & Caribbean vs. OECD, 2023.
Source: OECD/African Union Commission/African Tax Administration Forum, Revenue Statistics in Africa 2025.

Margaret Levi's concept of quasi-voluntary compliance offers one explanation: taxpayers comply at high, sustained rates substantially because they trust the state to spend revenue on services they value, which implies that visible public investment can precede and enable tax capacity rather than only follow from it. This is a plausible and influential account, consistent with China's and South Korea's own sequencing, but it is a theoretical framework rather than a causally established finding, and it sits alongside the same reverse-causality caution raised above: a state with more revenue can more easily fund the visible services that build compliance, so the relationship is plausibly circular rather than strictly sequential. This report treats early, visible investment in citizens as a defensible starting point for closing Africa's revenue gap without claiming it is the only mechanism the literature supports.

Where South Korea and China complicate a liberal-institutionalist reading. Neither country industrialized under competitive democratic politics during its fastest growth decades; both had unusually capable, directive states. Singapore is a more extreme version of the same pattern: a city-state whose technocratic bureaucracy achieved income convergence within a generation, though its small scale, single-city geography, and unusual founding conditions limit how far its experience generalizes to a continent of 54 states with an average population near 28 million. What these cases share, and what the data suggest matters more immediately than a state's democratic or authoritarian form, is administrative continuity: the practical capacity to hold an infrastructure or industrial strategy steady across multiple political cycles.

Continuity without authoritarianism: multi-year development budgets. South Korea and China could take policy continuity largely for granted during their takeoff decades: the same ruling order stayed in power long enough for literacy campaigns, grid buildouts, and industrial strategies to run to completion. Most African states cannot assume this. Where democratic institutions are still consolidating and electoral accountability is weak, a change of government often brings policy

reversal rather than refinement, and development spending is budgeted one year at a time and renegotiated with each new administration. Nigeria shows the cost. In 2011 the Presidential Projects Assessment Committee counted 11,886 abandoned federal and state government projects, with an estimated 7.78 trillion naira needed to complete them. The Ajaokuta Steel Complex, begun in the late 1970s as the anchor of Nigeria's industrialization, has absorbed billions of dollars without reaching sustained production.

The European Union offers a democratic model of continuity. Its Multiannual Financial Framework sets binding spending ceilings over seven years (five for the first framework, 1988–1992). Each annual budget must fit within those ceilings, and if member states fail to agree a new framework in time, the final year's ceilings carry over automatically rather than lapsing. Cohesion policy, which finances the convergence of the EU's poorer regions, received €351.8 billion in 2014–2020, about a third of the budget, and its ceiling is treated as a spending objective rather than a maximum. The framework is agreed unanimously by elected member-state governments with the consent of the directly elected European Parliament, yet it binds successive national governments to the same development commitments across their electoral cycles.

The lesson for African states is not to suspend democratic choice but to legislate a narrow set of foundational priorities (power, basic education, health, agricultural productivity, and industrial infrastructure) into multi-year frameworks that outlast any single administration, while leaving annual budgets free to vary around them. The trade-off is real: binding future governments reduces their flexibility, and a framework that is badly designed at the outset locks in its errors. Such frameworks therefore need scheduled, rules-based review rather than open-ended discretion.

The moral dimension: what if the money is stolen? A budget protected from electoral reversal is also a larger and more predictable target for theft. Nigeria's history makes the risk concrete. Switzerland returned roughly $700 million of funds stolen by the late General Sani Abacha in 2005–2006, and Transparency International notes that a significant share of that money remains unaccounted for. When Switzerland returned a further $322.5 million in 2017, it attached conditions: the funds went to a World Bank-supported cash-transfer programme for poor households, under World Bank monitoring and independent civil-society monitoring led by the Africa Network for Environment and Economic Justice (ANEEJ) through its MANTRA project. In an early monitoring round, 78.8% of households visited confirmed receiving payments. The arrangement was imperfect, but it was traceable, and it offers a template: development money ring-fenced for a defined purpose, disbursed against verifiable results, and monitored by parties independent of the government spending it.

Embezzlement is not only a financial loss. Every naira diverted from a power project or a school is a cost borne by people who will never see the infrastructure their taxes and borrowing paid for. It also erodes the trust on which the quasi-voluntary compliance argument above depends: citizens who watch one development budget disappear have little reason to fund the next. Integrity in public finance is therefore a development capability in its own right, not an administrative afterthought.

The state-failure counterargument. The state-capacity argument above has a significant survivorship-bias problem: it is built substantially from cases where state-directed strategy succeeded, and says less about the larger number of cases where it did not. Africa's own 1960s–1980s experience with state-led import-substitution industrialization, and comparable experiences in Latin America over the same period, generally failed to produce sustained manufacturing competitiveness, burdened economies with inefficient state enterprises, and in several cases ended in debt crises. The broader industrial-policy literature (surveyed by, among others, Pack and Saggi) finds a genuinely mixed record. Successful cases share identifiable features, namely export discipline, sunset clauses on protection, and competent, insulated bureaucracies, but these features have proven difficult to replicate deliberately, and the same interventions that succeeded in South Korea failed in comparable settings without them. This report's policy blueprint reflects that record by recommending industrial policy paired with measurable performance benchmarks and exit conditions, rather than open-ended state direction, and by treating the South Korea and China cases as evidence that capable, continuity-oriented state direction can work, not as evidence that state direction reliably works.

10. Africa's Demographic Opportunity

Sub-Saharan Africa's population is 41% under 15, against a 25% global average, and is projected to grow from about 1.24 billion in 2024 to about 2.1 billion by 2050; Africa as a whole is projected to reach about 2.5 billion.

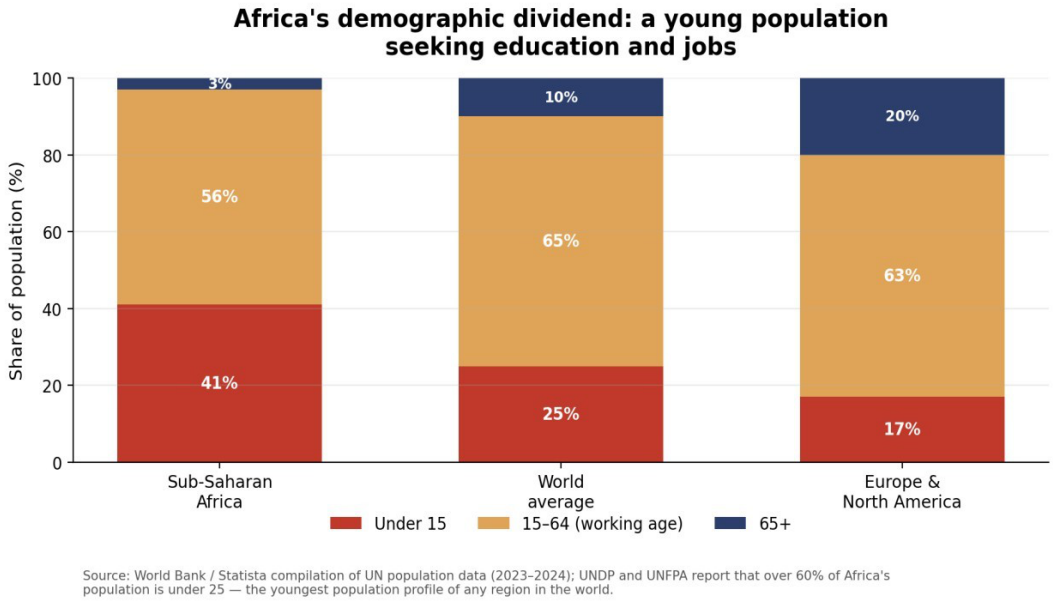


Figure 15. Population share by age band (%), Sub-Saharan Africa vs. World average vs. Europe & North America.

Source: World Bank / UN population data (2023–2024).

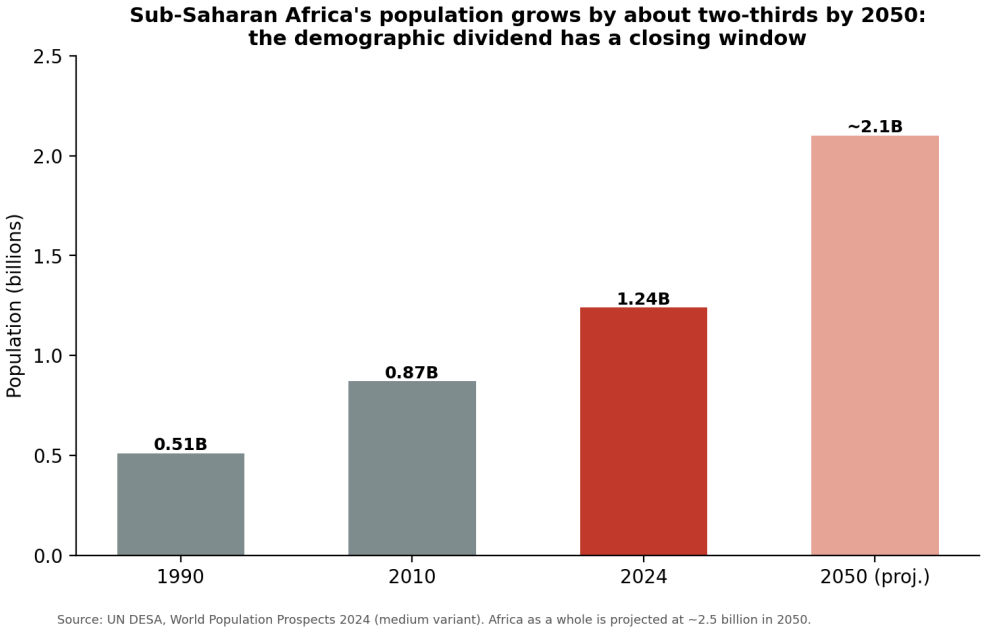


Figure 16. Sub-Saharan Africa population trajectory, 1990–2050 (projected).
Source: UN Department of Economic and Social Affairs, World Population Prospects 2024 (medium variant).

A youth bulge is a labor-market asset only if matched by education, health, and employment capacity; absent that, it is a source of labor-market pressure rather than a dividend, which is why this section should be read against Sections 2, 5, and 8 rather than in isolation. South Korea's and China's own industrialization occurred during comparable demographic windows, used deliberately to build labor-absorbing manufacturing capacity rather than treated as self-executing. A region adding roughly 850 million people by 2050, most entering working age over the next two decades, needs tens of millions of new formal or semi-formal jobs annually. On the evidence in Section 8, neither current informal-services growth nor manufacturing alone has yet demonstrated the capacity to absorb that scale, which is the central labor-market argument this report's later recommendations are built around.

11. Technology as a Cross-Cutting Capability

Sub-Saharan Africa's mobile subscription rate (93.4 per 100 people) is near-universal, but GSMA finds 63% of the population still does not use mobile internet.

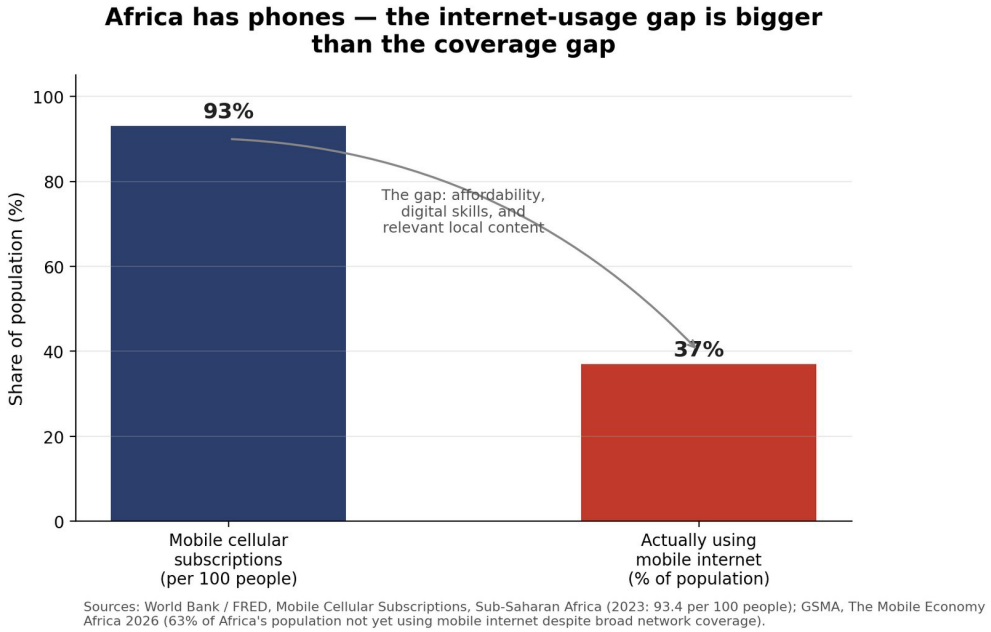


Figure 17. Mobile cellular subscriptions per 100 people vs. share of population actually using mobile internet, Sub-Saharan Africa / Africa, most recent available data.

Source: World Bank (2023); GSMA, The Mobile Economy Africa 2026.

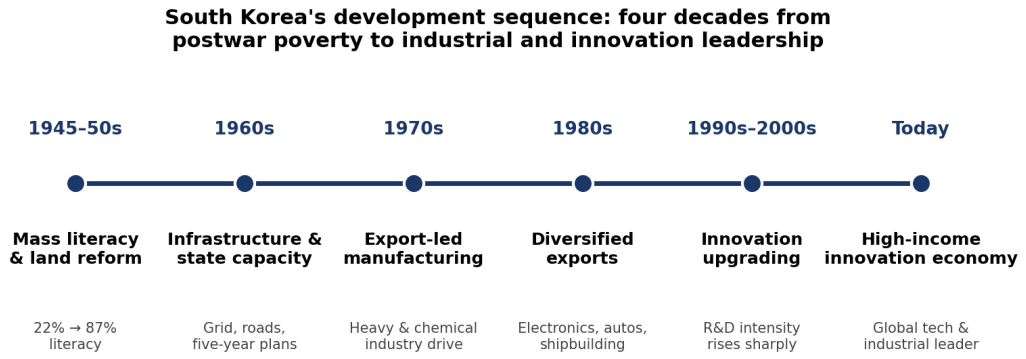
This gap, between infrastructure that exists and productive use that does not, is this report's clearest empirical illustration of the Nigeria puzzle introduced at the outset: a globally competitive technology sector coexisting with weak outcomes across most other capabilities. Where digital technology shows a well-evidenced growth effect (mobile money's expansion of financial inclusion, digital agricultural extension, telemedicine), it functions as a multiplier on activity or infrastructure that already exists, rather than as a generator of new activity on its own. This is consistent with, though does not by itself prove, this report's central proposition: technology strengthens prosperity when it is embedded within productive systems that already have some of the other capabilities in place, and its effect is limited where those systems are largely absent. Africa's technology sector, including Nigeria's, has produced genuine, globally competitive successes in fintech and logistics; the argument here is against treating that success as evidence that the other capabilities can be bypassed, not against the sector itself. Mobile phones do not replace electricity, fintech does not replace industry, and digital adoption does not replace the productive transformation the rest of this report examines.

12. A Wider Comparative Panel

Sections 2 through 11 built their sequence primarily from South Korea and China. This section tests how far that sequence generalizes using the wider comparator set introduced in Section 1, and engages directly with the two remaining counterarguments this report has not yet addressed: geography and resource dependence, and global trade fragmentation.

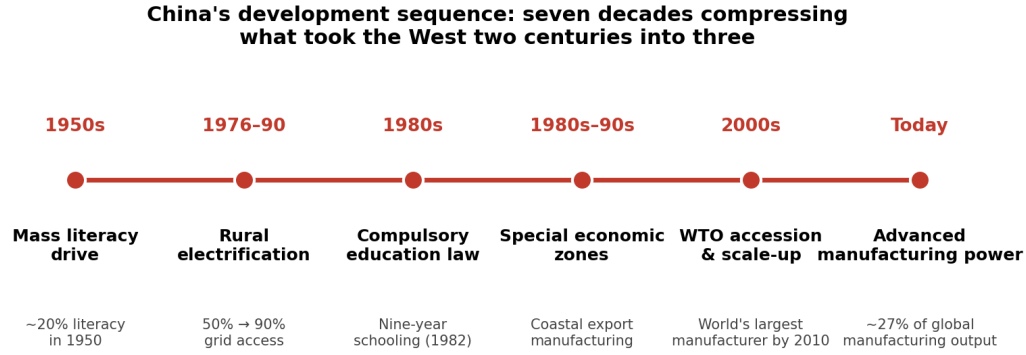
South Korea and China, briefly. South Korea's sequence ran from mass literacy (1945–1970), through infrastructure and state-capacity building (1960s) and export-led heavy industry (1970s)

financed substantially by directed foreign borrowing, to diversification into electronics and automobiles (1980s) and innovation-intensive upgrading (1990s–2000s). China's parallel sequence ran from mass literacy (1950s) and rural electrification (1976–1990), through Special Economic Zones and manufacturing FDI (1980s onward), to a WTO-era scale-up that made it the world's largest manufacturer by 2010. Both are shown in the figures below and summarized against Sub-Saharan Africa's current position in Table 2.



Source: World Bank; The Growth Lab at Harvard University, Atlas of Economic Complexity; literacy figures as cited in Section 2.

Figure 18. South Korea's development sequence, 1945–present.
Source: World Bank; The Growth Lab at Harvard University.



Source: World Bank; UN Statistics Division manufacturing data (2023); literacy and electrification figures as cited in Sections 2 and 4.

Figure 19. China's development sequence, 1950–present.
Source: World Bank; UN Statistics Division (2023).

Widening the panel. Measured by real GDP per capita growth since 2000, China (518%) leads a group including Vietnam (266%), India (235%), Bangladesh (208%), Romania (180%), and Poland (150%), several of which began closer to Sub-Saharan Africa's own 2000 starting point than South Korea's 1960s takeoff did.

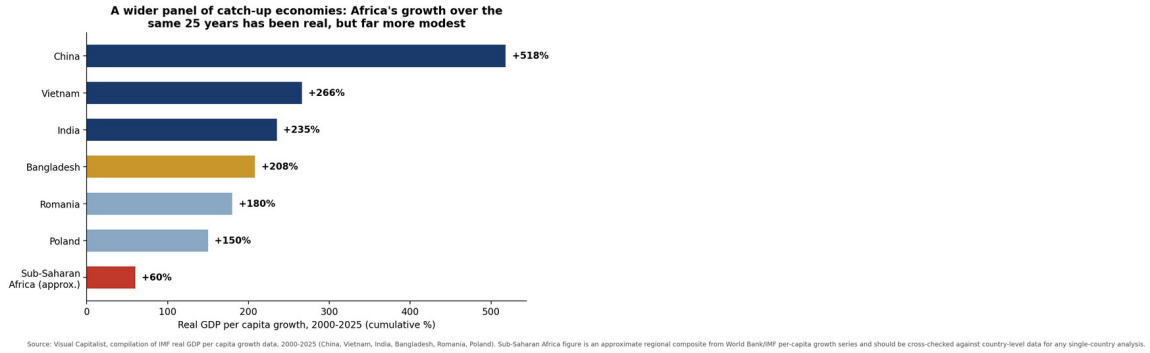


Figure 20. Real GDP per capita growth, 2000–2025 (cumulative %), selected catch-up economies vs. an approximate Sub-Saharan Africa composite.

Source: IMF data, 2000–2025, as compiled by Visual Capitalist. The Sub-Saharan Africa figure is an approximate composite.

Malaysia and Indonesia add a variant this report has not yet examined: both industrialized without abandoning commodity exports, using resource revenue (palm oil, rubber, and later electronics for Malaysia; energy and minerals for Indonesia) to fund the same human-capital and infrastructure investments China and Korea funded through manufacturing exports and directed borrowing. This suggests the underlying sequence tolerates more than one financing model. Poland shows the sequence does not require an authoritarian developmental state: since 1990 it has built human capital, integrated into EU capital and export markets, and industrialized within a liberal-democratic framework, and the IMF expects it to overtake several established Western European economies in per-capita income within years, a genuine complication of the state-capacity-over-democratic-form argument in Section 9. Bangladesh is the panel's clearest cautionary case: substantial manufacturing scale in a single sector (ready-made garments) produced real income growth but limited diversification and comparatively shallow productivity gains within the sector, illustrating that manufacturing entry is necessary but not sufficient without a deliberate push up the value chain.

Geography and resource dependence. A separate literature, associated with Jeffrey Sachs, argues that tropical disease burden, distance from coastal trade routes, and landlocked status are independent drags on growth that institutional or investment-based explanations understate. This report does not adjudicate that debate, but the resource-dependence evidence within it is directly relevant to several African economies. Oil-rich Nigeria and Angola have converted resource revenue into broad-based prosperity inconsistently; diamond-dependent Botswana has sustained among the world's highest growth rates since the 1960s; and Chile's copper-based development, sustained by comparatively strong institutions since its 1990s democratic transition, offers a non-African parallel. Mehlum, Moene, and Torvik's (2006) formal distinction between "producer-friendly" institutions, where resource rents are taxed and reinvested, and "grabber-friendly" institutions, where rents are captured by rent-seeking elites, is the best-supported explanation in the literature for this divergence. It implies that resource wealth is not itself the constraint facing Nigeria, Angola, or the DRC; institutional quality is, which returns the argument to Section 9 rather than away from it.

Global trade fragmentation. The clearest emerging challenge to this report's manufacturing-sequencing argument is not automation alone but the trading environment itself. UNCTAD's 2024 Trade and Development Report argues explicitly that "the traditional model of manufacturing-led export growth becomes less effective" as merchandise trade growth slows and geopolitical realignment reshapes supply chains. US-China bilateral trade fell roughly 30% between 2024 and 2025, with the reallocation concentrated in Vietnam, India, Indonesia, and Mexico rather than in Africa. The WTO's own gravity-model analysis finds only modest fragmentation along geopolitical lines to date and no clear evidence of broad regionalization, so the claim that trade fragmentation forecloses export-led manufacturing catch-up is not yet strongly supported empirically. But the direction of travel, and the fact that African economies are not currently among the fragmentation's beneficiary "connector" economies, is a genuine headwind against the South Korea and China sequence. This report's policy recommendations account for it explicitly, including through the emphasis on the African Continental Free Trade Area and regional integration in Section 13.

Table 2 summarizes South Korea, China, and Sub-Saharan Africa's current position across the indicators discussed in Sections 2–8.

Indicator	South Korea	China	Sub-Saharan Africa (latest)
Adult literacy	22% (1945) to 93% (~1988)	~20% (1950) to 97% (2020)	~68% (2023), rising
Life expectancy	Low 50s (1960s) to 83.4 yrs	~44 yrs (1960) to ~78–79 yrs	~62.7 yrs (2025)
Cereal yield	~6,300–6,800 kg/ha	6,380 kg/ha (2022)	~1,650 kg/ha (2020)
Manufacturing value added	~25% of GDP	~25% of GDP	~10% of GDP
Gross fixed capital formation	Above 30% of GDP through its takeoff decades	40.5% of GDP (2023)	25.3% of GDP (2023)
Foreign capital inflow (peak)	~10% of GDP (1971), predominantly loans	~5% of GDP (1990s), predominantly manufacturing FDI	~2% of GDP (1990–2022 avg.)

Table 2. South Korea and China's catch-up indicators against Sub-Saharan Africa's current position. Sources as cited in Sections 2–8.

13. Policy Blueprint

Each recommendation below is annotated with the strength of evidence behind it, following the methodology distinctions set out earlier in this report. “Strongly supported” indicates convergent descriptive, historical, and where available causal evidence; “well supported but contested” indicates a plausible, historically consistent case with genuine debate in the literature; “context-dependent” indicates a recommendation this report’s evidence suggests works only under specific, identifiable conditions.

The FutureLabs Prosperity Framework

The framework introduced in this report’s opening section is summarized in full below. It proposes that countries become prosperous not by maximizing a single variable, technology included, but by building a broad base of complementary capabilities that support industrialization, productivity growth, and structural transformation. Institutions and state capability form the foundation because Section 9 finds them the capability most consistently associated with a state’s ability to sustain investment in every other layer across political cycles. Education, health, agriculture, energy, infrastructure, capital formation, development finance, and manufacturing and tradable services build on that foundation in roughly the order the historical cases in this report accumulated them. Technology and innovation is deliberately not a stacked layer: it is a cross-cutting capability that strengthens whichever of the others a country has actually built, which is why Nigeria’s advanced fintech sector, discussed throughout this report, has not by itself produced the broad-based prosperity a fuller accumulation of capabilities would support.

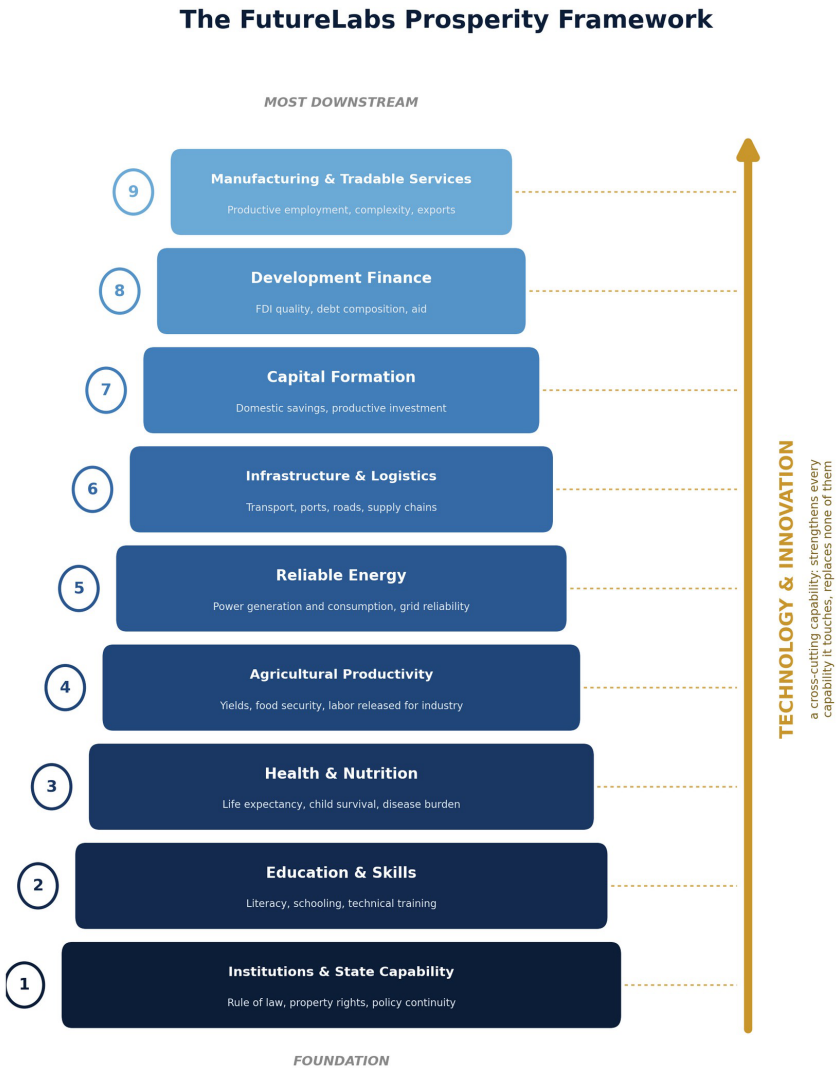


Figure 21. The FutureLabs Prosperity Framework.

FutureLabs Research original framework, synthesizing Sections 2–11. Each capability supports the ones above it; technology and innovation cross-cuts all of them, strengthening whichever capabilities a country has built and offering less where they are absent.

For governments

- **Strongly supported:** Treat literacy, basic education, and health-system capacity as a multi-decade commitment insulated from electoral cycles, following China's 1982 constitutional entrenchment of compulsory education. The descriptive and historical evidence for this is the most consistent in the report; no catch-up case examined here reached sustained growth without it.

- **Strongly supported:** Invest in agricultural extension, seed varieties, fertilizer access, and irrigation toward the South Asian yield benchmark. This follows directly from the labor-release mechanism documented in Section 5 and is one of the few recommendations in this report with reasonably strong causal evidence behind the underlying agronomic interventions, even though the macro growth effect remains historically inferred.
- **Well supported but contested:** Prioritize industrial- and commercial-grade grid capacity, not only household access. The consumption-versus-access distinction in Section 4 is descriptively clear; the causal magnitude of electrification's growth effect is more modest than commonly assumed, per the quasi-experimental literature cited there.
- **Context-dependent:** Pursue sector-targeted industrial policy for manufacturing, paired with export discipline, sunset clauses, and measurable performance benchmarks. Section 9's review of industrial-policy failures means this recommendation follows from the cases where such policy worked, not from a general claim that state-directed industrial policy is reliable; it should be adopted only where the state-capacity and continuity preconditions in Section 9 are plausibly present.
- **Well supported but contested:** Legislate multi-year development frameworks of five to seven years for a narrow set of foundational investments (power, basic education, health, agricultural productivity, and industrial infrastructure), with binding ceilings that carry over automatically if a new framework is not agreed, following the EU's Multiannual Financial Framework. Pair them with scheduled, rules-based review, and require that abandoned projects be completed or formally closed before new ones of the same type begin (Section 9).
- **Well supported but contested:** Ring-fence development funds and disburse them against verifiable milestones, with independent audit, civil-society monitoring, and public project-level reporting of spending and completion, following the monitored return of Abacha funds in 2017. The evidence is case-based rather than systematic, but the cost of the alternative is documented (Section 9).
- **Well supported but contested:** Sequence revenue mobilization behind visible public investment rather than enforcement alone, consistent with quasi-voluntary compliance theory, while recognizing the reverse-causality caution raised in Section 9.
- **Context-dependent:** Where a country's skill base and comparative advantage support it, invest in tradable knowledge-intensive services alongside, not instead of, manufacturing strategy, following the Costa Rica and India evidence in Section 8.

For the private sector

- Treat manufacturing and light-industrial investment as a multi-decade asset class; treat digital tools as complements to existing informal activity and infrastructure rather than substitutes for it (well supported, Sections 8 and 11).

- Invest directly in vocational and technical training pipelines rather than assuming public systems will supply industrial-ready labor on an investment's timeline (context-dependent on sector and country).

For universities and development partners

- Expand engineering, agronomy, public health, and vocational-technical programs, and build applied research partnerships with domestic manufacturing, agriculture, and health systems (well supported by the human-capital evidence in Sections 2–3 and 5).
- Shift financing toward productive electrification, agricultural productivity, and health-system capacity alongside continued household-access programs; favor grant and equity-like instruments over further sovereign lending where debt service is already elevated (well supported by Sections 4, 5, and 7).
- Tie multi-year financing commitments to independent monitoring and results-based disbursement, so that predictable funding does not become a predictable target (Section 9).
- Pair tax-administration support with support for the citizen-facing services that make that capacity politically sustainable (context-dependent, per the sequencing debate in Section 9).

For investors

- Distinguish countries where digital adoption outpaces foundational investment from those where it is layered on genuine gains in electrification, agriculture, and manufacturing or tradable services (well supported by this report's core argument).
- Weight state capacity, policy continuity, and debt-composition quality alongside headline growth indices in country-risk assessment (well supported by Sections 7 and 9).
- Favor markets showing evidence of climbing the value chain over single-sector concentration, per the Bangladesh caution in Section 12 (context-dependent).

Conclusion

The evidence reviewed in this report suggests that prosperity is not downloaded; it is built. Across the successful economies examined, prosperity emerged from the accumulation of complementary capabilities (education, health, agriculture, energy, infrastructure, capital formation, development finance, institutions, and manufacturing or tradable services) that supported industrialization, productivity growth, and structural transformation over time. Technology strengthened these processes in every case where it contributed meaningfully to growth. It did not replace them in any case this report examined.

Stated plainly, the data in this report point one way. Software cannot build a power grid, a factory, or a functioning ministry. Every economy examined here that converged with richer countries first built a productive base of literacy, health, power, agricultural productivity, and industrial capacity, and

technology then multiplied what that base could do. Industrialization came first in every large economy. Even India's services boom, the strongest alternative, has left it at less than half of China's income per person, and Costa Rica's services rose on an electronics-manufacturing base. Nigeria shows what happens when technology arrives first: a world-class fintech sector on top of an economy still short of power, yields, and factories.

The gaps documented in this report (in literacy, electricity, cereal yields, manufacturing, investment, and tax capacity) are therefore where development budgets should go. Because closing them takes decades, the budgets that fund them must outlast electoral cycles, as the EU's multi-year framework does for Europe's poorer regions. And because a budget that outlasts elections is also a larger target for theft, it must be protected by ring-fencing, independent monitoring, and public reporting. Long horizons, the right priorities, and integrity in spending: in South Korea and China, a directive state supplied the first; in Africa's democracies, it will have to be written into law.

For Africa, and particularly for Nigeria, the challenge this report identifies is therefore not choosing between technology and traditional development. It is building the broad foundation of complementary capabilities that allows innovation, industry, and enterprise, technology among them, to translate into the kind of widespread, sustained prosperity that a competitive technology sector has not, on its own, been sufficient to deliver anywhere examined in this report. That foundation is built differently in Lagos than in Kigali, and differently again in Nairobi or Gaborone; the country-specific analysis this report repeatedly calls for, rather than a single continental prescription, is where its evidence should ultimately be applied.

Methodology, Evidence, and Limitations

This report combines descriptive statistics, historical case comparison, selectively cited causal-inference literature, and normative policy interpretation, as distinguished in the Note on Method above. Four further limitations bear on how its findings should be used.

Selection and generalizability. South Korea and China are extreme, not typical, catch-up cases, selected partly because they succeeded. The wider panel in Section 12 mitigates but does not eliminate this bias, since Vietnam, Poland, Malaysia, Costa Rica, and Singapore were also selected for their comparative success. All benefited from historically specific circumstances (Cold War strategic investment, WTO-era trade openness, EU accession) that will not replicate identically for African economies today, and the report's recommendations should be read as evidence about mechanism rather than as a claim that any single trajectory is directly replicable under 2026 conditions.

Causation and correlation. Where this report describes a relationship as causal, it cites literature using experimental, quasi-experimental, or formal-theoretic identification (Mehlum, Moene, and Torvik on the resource curse; the electrification literature in Section 4). Elsewhere, including the central relationships between institutions and income, and between tax capacity and public investment, the report describes robust cross-country association with a plausible, contested mechanism, and states explicitly, per Besley and Persson's formal treatment, that causation plausibly

runs in both directions simultaneously.

Heterogeneity. Continental and Sub-Saharan averages mask substantial country-level variation, addressed directly in Section 1's typology; a policymaker or investor should treat this report's aggregate figures as a starting frame, not a substitute for country-specific analysis.

Contested and evolving evidence. The automation-and-manufacturing, services-led-development, institutional-sequencing, and trade-fragmentation debates engaged in Sections 8, 9, and 12 are active areas of research as of 2026, drawing on literature published as recently as 2025 (Diao, Ellis, McMillan, and Rodrik on Africa-specific manufacturing productivity) and 2026 (UNCTAD and WTO trade-fragmentation data). This report's treatment of them reflects the current state of an unsettled literature rather than a resolved consensus, and should be revisited as that literature develops.

Data Sources

This report draws on World Bank World Development Indicators and Worldwide Governance Indicators; World Bank/ESMAP Tracking SDG7; the IEA's SDG7 Data and Projections; OECD/AUC/ATAF Revenue Statistics in Africa 2025; WHO and UN IGME child mortality data; FAO/FAOSTAT and the World Bank's Africa's Pulse series; UN DESA World Population Prospects; UNDP and UNFPA demographic reporting; The Growth Lab at Harvard University's Atlas of Economic Complexity; the Maddison Project Database; the ISS African Futures programme; GSMA's Mobile Economy Africa series; the African Development Bank; BloombergNEF's Africa Power Transition Factbook; Our World in Data; UNCTAD's External Debt Sustainability and Development report and Trade and Development Report 2024; the WTO's fragmentation research (2023–2024); the IMF; the NBER; the ODI; Acemoglu and Robinson (2012); Besley and Persson (2011); Mehlum, Moene, and Torvik (2006); Levi's work on quasi-voluntary compliance; Rodrik's work on premature deindustrialization; Diao, Ellis, McMillan, and Rodrik (2025); and Hallward-Driemeier and colleagues' World Bank research on services-led development; Council Regulation (EU, Euratom) 2020/2093 and European Parliament research on the EU Multiannual Financial Framework; the Nigerian Presidential Projects Assessment Committee report (2011); Transparency International; and ANEEJ's MANTRA monitoring reports on the returned Abacha funds, among other sources cited at point of use.

Historical GDP-per-capita figures before roughly 1950 are drawn from the Maddison Project's long-run reconstructions and should be treated as estimates. Where a figure could not be verified from an authoritative current source, this report says so explicitly rather than estimating with false precision. All references to "current" or "today" figures reflect the most recently published data available as of July 2026, even where the underlying survey year is 2023 or 2024.

This report was prepared by Udim Manasseh, FutureLabs Research, for governments, development agencies, investors, and researchers assessing the drivers of long-run African economic growth. Its findings should be read alongside country-specific analysis before informing individual policy or investment decisions.